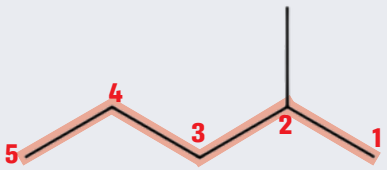
CARD A01 STRUCTURE ⟶ NAME

A01 FOUNDATION C6H14

Numbered Parent



Naming Trail

Parent: Longest chain = 5C —> pentane. Several 5C path exist. Only two of them give the same name.

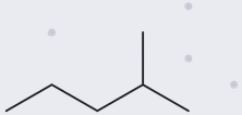
Numbering: Right —> left gives 4; left —> right gives 2. Therefore the parent chain is numbered from right.

Substituents: methyl on C2 —> 2-methyl

My Name

2-methylpentane

The Alkane Check



2-methylpentane

Returned name: 2-methylpentane

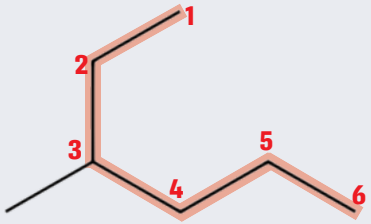
Match

☑ longest chain confirmed ☑ lowest locant ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A02 STRUCTURE ⟶ NAME

A02 FOUNDATION **C7H16**

Numbered Parent



Naming Trail

Parent: Longest chain = 6C —> hexane.

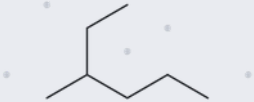
Numbering: Right —> left gives 4; left —> right gives 3. Therefore the parent chain is numbered from left.

Substituents: methyl on C3 —> 3-methyl

My Name

3-methylhexane

The Alkane Check



3-methylhexane

Returned name: 3-methylhexane

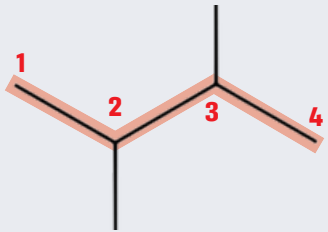
Match

☑ longest chain confirmed ☑ lowest locant ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A03 STRUCTURE ⟶ NAME

A03 FOUNDATION C6H14

Numbered Parent



Naming Trail

Parent: Longest chain = 4C —> butane. Several 4C path exist but each carries two methyl groups, so they lead to the same name.

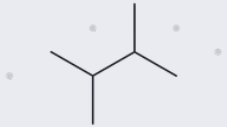
Numbering: Right —> left gives 2,3; left —> right gives 2,3. Regardless of the numbering, they yield the same locant for substituents.

Substituents: methyl on C2, C3 —> 2,3-dimethyl

My Name

2,3-dimethylbutane

The Alkane Check



2,3-dimethylbutane

Returned name: 2,3-dimethylbutane

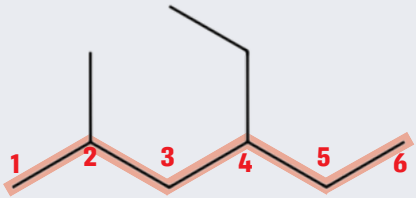
Match

☑ longest chain confirmed ☑ lowest locant ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A04 STRUCTURE ⟶ NAME

A04 DEVELOPING C9H2O

Numbered Parent



Naming Trail

Parent: Longest chain = 6C —> hexane. Several 6C path exist but each carries two substituents (methyl, ethyl), so they lead to the same name.

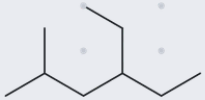
Numbering: Right —> left gives 3,5; left —> right gives 2,4. First point difference 2<3, so numbered from left.

Substituents: methyl on C2; ethyl on C4 —> 4-ethyl-2-methyl

My Name

4-ethyl-2-methylhexane

The Alkane Check



4-ethyl-2-methylhexane

Returned name: 4-ethyl-2-methylhexane

Match

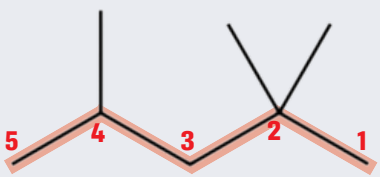
☑ longest chain confirmed ☑ lowest locant ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A05 — STRUCTURE ⟶ NAME

A05 DEVELOPING

C8H18

Numbered Parent



Naming Trail

Parent: Longest chain = 5C —> pentane. Several 5C path exist but each carries three methyl substituents so they lead to the same name.

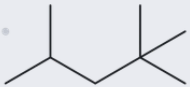
Numbering: Right —> left gives 2,2,4; left —> right gives 2,4,4. First point difference 2<4, so numbered from right.

Substituents: methyl on C2, C2, C4 —> 2,2,4-trimethylpentane

My Name

2,2,4-trimethylpentane

The Alkane Check



2,2,4-trimethylpentane

Returned name: 2,2,4-trimethylpentane

Match

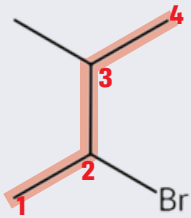
- ☑ longest chain confirmed
- ☑ lowest locant
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

CARD A06 — STRUCTURE ⟶ NAME

A06 DEVELOPING

C5H11Br

Numbered Parent



Naming Trail

Parent: Longest chain = 4C —> butane. Several 4C path exist but each carries the same substituents (methyl and bromine) so they lead to the same name.

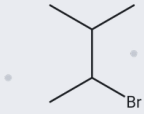
Numbering: The substituents are equal in position (both from right and left at 2, 3). Alphabetically, the chain was numbered from the left .

Substituents: bromine on C2; methyl on C3 —> 2-bromo-3-methyl

My Name

2-bromo-3-methylbutane

The Alkane Check



2-bromo-3-methylbutane

Returned name: 2-bromo-3-methylbutane

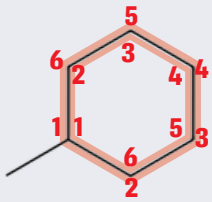
Match

☑ longest chain confirmed ☑ lowest locant ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A07 — STRUCTURE ⟶ NAME

A07 DEVELOPING C7H14

Numbered Parent



Naming Trail

Parent: Longest chain = 6C —> cyclohexane. Since a ring is present and no other carbon chain is present, the ring is the parent chain.

Numbering: The numbering must start from the only substituent. But at any position on the ring, the methyl is always on C1 so numbering don't matter.

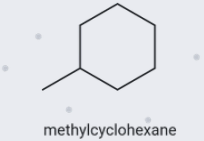
Substituents: methyl on C1 —> 1-methyl or just methyl since at any point it's at C1

My Name

✗ methylhexane

✓ methylcyclohexane

The Alkane Check



methylcyclohexane

Returned name: methylcyclohexane

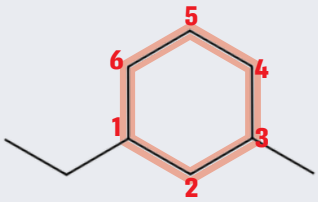
✗ **Not Match**, prefix "cyclo" was accidentally ommited

✓ longest chain confirmed ✓ lowest locant ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A08 — STRUCTURE ⟶ NAME

A08 CHALLENGE **C9H18**

Numbered Parent



Naming Trail

Parent: Longest chain = 6C —> cyclohexane. Since the ring is the longest carbon chain present, the ring is the parent chain.

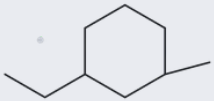
Numbering: The substituents are equal in positions (both from left and right at 1, 3). Alphabetically, the chain was numbered from the left.

Substituents: ethyl on C1; methyl on C3 —> 1-ethyl-3-methyl

My Name

1-ethyl-3-methylcyclohexane

The Alkane Check



1-ethyl-3-methylcyclohexane

Returned name: 1-ethyl-3-methylcyclohexane

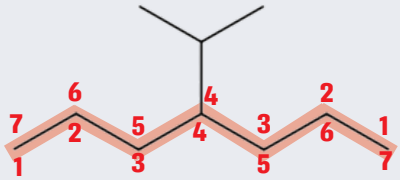
✓ **Match**

☑ longest chain confirmed ☑ lowest locant ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A09 — STRUCTURE ⟶ NAME

A09 CHALLENGE **C10H22**

Numbered Parent



Naming Trail

Parent: Longest chain = 7C —> heptane.

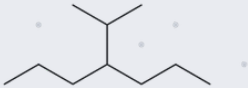
Numbering: Right —> left gives 4; left —> right gives 4. Regardless of the numbering, they yield the same locant for the substituent.

Substituents: isopropyl on C4 —> 4-(propan-2-yl)

My Name

4-(propan-2-yl)heptane

The Alkane Check



2-methyl-3-propylhexane

Returned name: 2-methyl-3-propylhexane

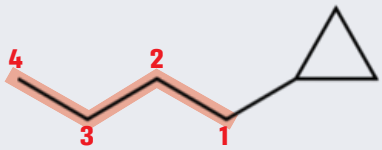
✗ **Not Match**, alkane used a shorter chain "hexane"

☑ longest chain confirmed ☑ lowest locant ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A10 — STRUCTURE ⟶ NAME

A10 CHALLENGE C7H14

Numbered Parent



Naming Trail

Parent: Longest chain = 4C —> butane. Although a present, it only contains 3C that is 1 carbon shorter than the other chain.

Numbering: Right —> left gives 1; left —> right gives 4. Therefore the parent chain is numbered from right.

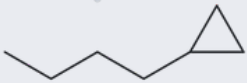
Substituents: cyclopropyl on C1 —>1-cyclopropyl

My Name

✗ 1-cyclopropylbutane

✓ butylcyclopropane

The Alkane Check



butylcyclopropane

Returned name: butylcyclopropane

✗ **Not Match** , the ring "cyclopropane" must be the parent regardless of carbon count

☑ longest chain confirmed ☑ lowest locant ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD B01 **NAME → STRUCTURE**

B01 FOUNDATION **2-methylpropane**

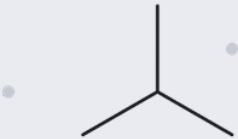
Name Decoded

Parent: propane —> a 3-carbon chain

Substituents: methyl on C2

Expected formula: 3 + 1 = 4C; 7 + 3 = 10H —> C4H10

Bond Line Structure



Formula Check

Counted from the drawing: C4H10

Back-Check

Returned by the alkane: 2-methylpropane

✓ **Same as the given name**

☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

CARD B02 NAME → STRUCTURE

B02 FOUNDATION 3-ethylpentane

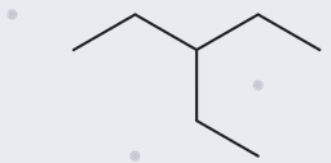
Name Decoded

Parent: pentane —> a 5-carbon chain

Substituents: ethyl on C3

Expected formula: 5 + 2 = 7C; 11 + 5 = 16H —> C7H16

Bond Line Structure



Formula Check

Counted from the drawing: C7H16

Back-Check

Returned by the alkane: 3-ethylpentane

✓ **Same as the given name**

☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

CARD B03 NAME ⟶ STRUCTURE

B03 FOUNDATION 2,2-dimethylbutane

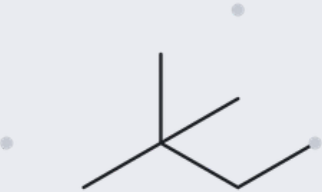
Name Decoded

Parent: butane —> a 4-carbon chain

Substituents: two methyl on C2

Expected formula: 4 + 2 = 6C; 8 + 6 = 14H —> C6H14

Bond Line Structure



Formula Check

Counted from the drawing: C6H14

Back-Check

Returned by the alkane: 2,2-dimethylbutane

✓ **Same as the given name**

☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

CARD B04 NAME → STRUCTURE

B04 DEVELOPING 3-ethyl-2-methylpentane

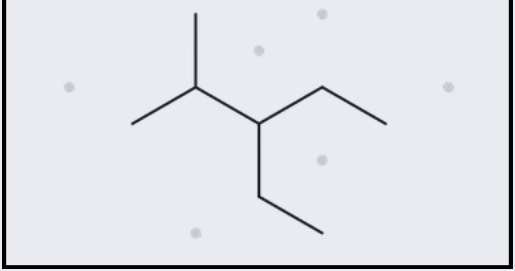
Name Decoded

Parent: pentane —> a 5-carbon chain

Substituents: methyl on C2; ethyl on C3

Expected formula: 5 + 1 + 2 = 8C; 10 + 3 + 5 = 18H
—> C8H18

Bond Line Structure



Formula Check

Counted from the drawing: C8H18

Back-Check

Returned by the alkane: 3-ethyl-2-methylpentane

✓ **Same as the given name**

☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

CARD B05 NAME ⟶ STRUCTURE

B05 DEVELOPING 1-choloro-2-methylpropane

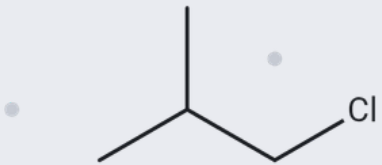
Name Decoded

Parent: propane —> a 3-carbon chain

Substituents: methyl on C2; chlorine on C1

Expected formula: 3 + 1 = 4C; 6 + 4 = 10H —> C4H10; Cl replaces one H —> C4H9Cl

Bond Line Structure



Formula Check

Counted from the drawing: C4H9Cl

Back-Check

Returned by the alkane: 1-chloro-2-methylpropane

✓ **Same as the given name**

☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

CARD B06 NAME → STRUCTURE

B06 DEVELOPING 1,2-dimethylcyclopentane

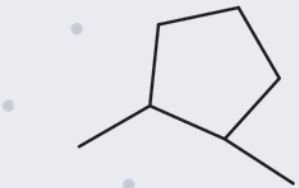
Name Decoded

Parent: cyclopentane —> a 5-carbon ring

Substituents: methyl on C1, C2

Expected formula: 5 + 1 + 1 = 7C; 8 + 3+ 3 = 14H
—> C7H14

Bond Line Structure



Formula Check

Counted from the drawing: C7H14

Back-Check

Returned by the alkane: 1,2-dimethylcyclopentane

✓ **Same as the given name**

☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

CARD B07 NAME → STRUCTURE

B07 DEVELOPING 4-ethyl-3,3-dimethylheptane

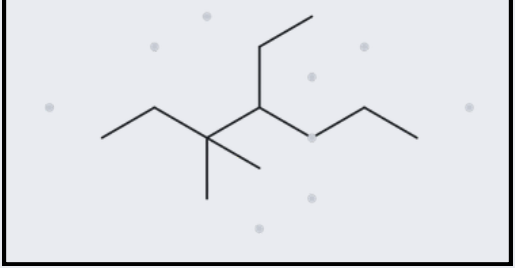
Name Decoded

Parent: heptane —> a 7-carbon chain

Substituents: two methyl on C3; one ethyl on C4

Expected formula: 7 + 2 + 2 = 11C; 13 + 6+ 5 = 24H
—> C11H24

Bond Line Structure



Formula Check

Counted from the drawing: C11H24

Back-Check

Returned by the alkane: 4-ethyl-3,3-dimethylheptane

✓ **Same as the given name**

☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

CARD B08 **NAME ⟶ STRUCTURE**

B08 CHALLENGE **tert-butylcyclohexane**

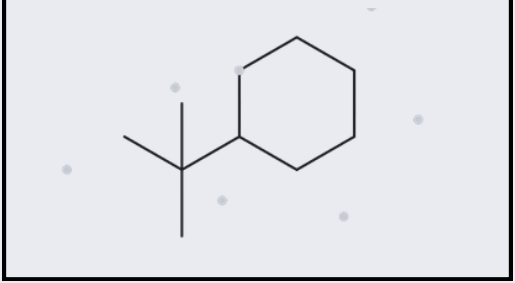
Name Decoded

Parent: cyclohexane —> a 6-carbon ring

Substituents: tert-butyl on C1

Expected formula: 6 + 4 = 10C; 11 + 9 = 20H
—> C10H20

Bond Line Structure



Formula Check

Counted from the drawing: C10H20

Back-Check

Returned by the alkane: (1,1-dimethylethyl)cyclohexane

✓ **Same as the given name** (written in alkane format)

- ☑ parent length
- ☑ substituent positions
- ☑ formula agrees
- ☑ back-check recorded

CARD B09 **NAME → STRUCTURE**

B09 CHALLENGE **3-ethyl-1,1-dimethylcyclohexane**

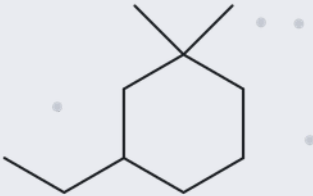
Name Decoded

Parent: cyclohexane —> a 6-carbon ring

Substituents: two methyl on C1; one ethyl on C3

Expected formula: 6 + 2 + 2 = 10C; 9 + 6+ 5 = 20H
—> C10H20

Bond Line Structure



Formula Check

Counted from the drawing: C10H20

Back-Check

Returned by the alkane:
3-ethyl-1,1-dimethylcyclohexane

✓ **Same as the given name**

☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded

CARD B10 **NAME → STRUCTURE**

B10 CHALLENGE **2-ethylbutane**

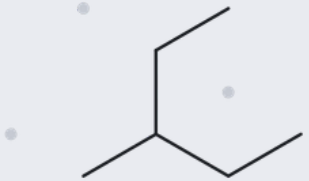
Name Decoded

Parent: butane —> a 4-carbon chain

Substituents: ethyl on C2

Expected formula: 4 + 2 = 6C; 9 + 5 = 14H
—> C6H14

Bond Line Structure



Formula Check

Counted from the drawing: C6H14

Back-Check

Returned by the alkane: 3-methylpentane

☒ **Not the same as given the name**

there is a longest carbon chain is 5 not 4, pentane instead of butane

☑ parent length ☑ substituent positions ☑ formula agrees ☑ back-check recorded